Keywords: safe reinforcement learning • constrained MDP • representation learning • low-rank MDP

Primal-Dual Representation Learning for Low-Rank Constrained MDPs

The First Sample-Efficient Algorithms with Theoretical Guarantees for Continuous-State Safe RL Under Both Soft and Hard Violation Constraints

By Chenhao Zhou, Chao Zhang, Wensong Bai, Hanbin Zhao, Hui Qian

NeurIPS 2026 (Poster #151600)

Safe reinforcement learning requires an agent to maximize return while keeping constraint violations within a budget—a challenge that, for continuous high-dimensional state spaces, has lacked end-to-end theoretical guarantees when the transition model must be learned from data. This paper introduces REP-PD-TEG and REP-PD-H-TEG, two primal-dual algorithms that jointly learn low-rank transition representations and satisfy safety constraints, achieving $\tilde{O}(\sqrt{K})$ return regret (soft violation) and certified per-episode constraint satisfaction (hard violation) respectively. The key insight is that representation learning and dual updates can be decoupled: the former proceeds by maximum-likelihood exploration, independent of the safety objective, while the latter handles constraints via an optimistic Lagrangian with a stabilized dual search.

AT A GLANCE

Problem: Safe RL in continuous state spaces requires learning transition representations and satisfying constraints simultaneously; no prior method provided end-to-end guarantees for both.

Why it matters: Real-world safe RL (robotics, autonomous driving) has continuous state spaces; tabular guarantees do not apply, and deep safe RL methods (CPO, PDO) lack theoretical backing.

Method: Decoupled representation learning via ML exploration, followed by optimistic Lagrangian planning (REP-PD-TEG for soft violation) or per-episode feasibility certification (REP-PD-H-TEG).

Result: $\tilde{O}(\sqrt{K})$ regret and $\tilde{O}(K^{3/4})$ total samples (soft); certified per-episode zero violation (hard).

Evidence: Continuous-state safety gridworld, constrained LQR; ablations on features, exploration rate, dual step size.

The Big Picture

Safe RL is the problem of training policies that maximize cumulative reward while respecting safety constraints expressed as budgets on cumulative or per-episode costs. It is central to deploying RL in physical systems—robots that should not fall, vehicles that should not collide, and medical dosing agents that should not exceed toxic thresholds.

The theoretical safe RL literature has largely studied tabular settings, where finite-sample guarantees can be derived from counting arguments over a discrete state space. For continuous state spaces, function approximation is necessary, and the low-rank MDP model—where the transition kernel factorizes as $P(s' \mid s, a) = \phi(s, a)^\top \mu(s')$ for low-dimensional features ϕ and μ—is the natural extension that retains mathematical tractability.

Until this paper, no work had provided end-to-end sample-efficiency guarantees for the combined problem of learning the representation ϕ and satisfying a safety constraint in a continuous-state low-rank CMDP.

Why Existing Approaches Fall Short

Tabular constrained RL (C-UCBVI, OptPess-LP): Provides finite-sample guarantees but requires discretizing the state space, which is computationally intractable in high dimensions and introduces approximation error proportional to the discretization resolution.

Linear function approximation for CMDPs: Assumes the true features ϕ are given; cannot handle the setting where they must be estimated from data, which introduces feature estimation error that can violate constraint guarantees.

Deep safe RL (CPO, PDO, RCPO): Achieves strong empirical performance but provides no finite-sample bounds on regret or constraint violation; convergence guarantees require idealized assumptions rarely met in practice.

Unconstrained low-rank MDP representation learning: Techniques such as FLAMBE and REPUCB learn ϕ by maximum likelihood and achieve provably near-optimal policies—but the dual variable of the safety constraint interacts non-trivially with the UCB bonus, breaking the direct extension.

Core Mechanism

The key architectural insight is that representation learning and safety constraint handling can be *decoupled*:

1. **Representation learning:** Run a decaying epsilon-greedy exploration policy to collect transition data; fit $\hat{\phi}$ by maximum-likelihood. This is independent of the safety objective.

2. **Primal-dual planning:** Use $\hat{\phi}$ to construct UCB-style optimistic estimates of both the return $\hat{V}_\pi^r$ and the constraint utility $\hat{V}_\pi^c$. Plan with the optimistic Lagrangian: $\max_\pi \min_\lambda \hat{V}_\pi^r - \lambda(\hat{V}_\pi^c - b)$. Update λ by dual gradient ascent.

REP-PD-TEG handles the soft (cumulative) violation case with standard UCB bonuses and gradient ascent on the dual.

REP-PD-H-TEG handles the hard (per-episode certified) violation case by adding a lower-confidence constraint certificate: before deploying any policy, the algorithm verifies that the pessimistic constraint estimate $\underline{V}_\pi^c$ is below the budget b, preventing episodes that violate the constraint even in the worst case.

Technical Formulation

The low-rank CMDP structure assumes:

$$P(s' \mid s, a) = \phi(s, a)^\top \mu(s'), \quad \phi : \mathcal{S} \times \mathcal{A} \to \mathbb{R}^d$$

The constrained optimization problem is:

$$\max_\pi V_\pi^r \quad \text{s.t.} \quad V_\pi^c \le b$$

The Lagrangian relaxation is:

$$\mathcal{L}(\pi, \lambda) = V_\pi^r - \lambda(V_\pi^c - b), \quad \lambda \ge 0$$

Maximum-likelihood feature learning from an epsilon-greedy exploration policy with budget T_0 achieves feature error $\|\hat{\phi} - \phi^*\|_\infty \le \epsilon_\phi$ with high probability for $T_0 = \tilde{O}(d^2/\epsilon_\phi^2)$.

With the learned features, the optimistic return and constraint estimates are:

$$\hat{V}_\pi^r = V_\pi^r(\hat{\phi}) + \underbrace{\beta_r \cdot b_\pi^\phi}_{\text{UCB bonus}}, \quad \hat{V}_\pi^c = V_\pi^c(\hat{\phi}) + \beta_c \cdot b_\pi^\phi$$

The main theoretical results for REP-PD-TEG are:

$$\text{Return regret} = \tilde{O}(\sqrt{K}), \quad \text{Total samples} = \tilde{O}(K^{3/4})$$

For REP-PD-H-TEG, the per-episode hard-violation guarantee holds with probability $1 - \delta$ for all K episodes simultaneously.

Learning or Inference Procedure

REP-PD-TEG:

1. Collect T_0 episodes of epsilon-greedy data; fit $\hat{\phi}$ by MLE.
2. For each episode k: compute optimistic $\hat{V}^r$, $\hat{V}^c$ using $\hat{\phi}$ and UCB bonuses; solve the optimistic Lagrangian for π_k; deploy π_k; update λ by gradient ascent: $\lambda \leftarrow \lambda + \eta_\lambda(V_{\pi_k}^c - b)$.

REP-PD-H-TEG:

1. Steps 1–2 as above, but before deploying π_k, compute the pessimistic lower bound $\underline{V}_{\pi_k}^c$ using a lower-confidence interval on $\hat{V}^c$.
2. If $\underline{V}_{\pi_k}^c \le b$: deploy π_k (certified safe).
3. Otherwise: invoke the stabilized dual search to find a feasible policy (shrink dual step size if the certificate is tight).

What the Guarantee Says

Soft violation (REP-PD-TEG): The cumulative return regret over K episodes is $\text{Reg}_K = \tilde{O}(\sqrt{K})$, and the total cumulative constraint violation is bounded by $\tilde{O}(\sqrt{K})$. Total episodes including exploration: $\tilde{O}(K^{3/4})$.

Hard violation (REP-PD-H-TEG): With probability $1 - \delta$, the deployed policy satisfies $V_{\pi_k}^c \le b$ for all $k = 1, \ldots, K$ simultaneously. Return regret is slightly worse than the soft case due to the conservative certification step but remains poly(K)-sublinear.

First result of its kind: The authors claim these are the first finite-sample guarantees for representation learning in low-rank CMDPs with continuous state spaces under both soft and certified hard violation settings.

Experimental Findings

- **Continuous safety gridworld:** REP-PD-TEG keeps cumulative constraint violations below budget; REP-PD-H-TEG achieves zero per-episode violations after an initial feature-learning phase.
- **Constrained LQR:** Both algorithms outperform CPO and PDO (which use fixed features) when the provided features are incorrect; performance matches CPO with correct features.
- **Benefit of learned features:** Methods with fixed (incorrect) features consistently violate the constraint budget; the adaptive feature learning of REP-PD-TEG removes this failure mode.

Ablations and Interpretation

Fixed vs. learned features: Ground-truth features in the primal-dual step achieve near-optimal performance; incorrect fixed features fail to satisfy constraints and achieve poor return, confirming the necessity of end-to-end representation learning.

Epsilon-greedy schedule: A decaying schedule is critical; a constant rate leads to over-exploration and slower convergence; an overly fast decay leads to poor feature quality and constraint violations.

Dual step size stability (REP-PD-H-TEG): The stabilized dual search (reduced step size near the con-

straint boundary) is essential for hard-violation safety; naive gradient ascent on the dual oscillates and produces intermittent per-episode violations.

Reference

Chenhao Zhou, Chao Zhang, Wensong Bai, Hanbin Zhao, Hui Qian. **Primal-Dual Representation Learning for Low-Rank Constrained MDPs**. NeurIPS 2026 (Poster #151600). https://neurips.cc/virtual/2026/poster/151600